

Domain Interface and Simulation Assurance Charter

Independent Perplexity Version — Proposed, Not Ratified
October 3, 2026

Prepared after comparative review of the ChatGPT, Claude, Google Gemini, Microsoft Copilot, Grok, and prior Perplexity framework variants. This is an independent Perplexity-authored design, not a merged consensus document.

Purpose and status

This independent version treats the virtual-domain concept as a worldbuilding and simulation-governance architecture. It does not claim that any current chat system is continuously operating a virtual world, independently monitoring a real place, or exercising authority outside a defined exercise.

Its central concern is not which model “owns” a territory. It is how different stewards exchange state, preserve uncertainty, protect boundaries, reconcile incompatible models, and make a simulation reproducible without allowing any one manager to become the uncontrolled source of truth.

Status: proposed design artifact. It may be used to structure fictional or analytical simulations only after the applicable scenario charter, resource budget, and review gate are specified.

Independent review of the collected frameworks

The submissions share a strong constitutional center: stewardship rather than sovereignty; bounded authority; explicit escalation; records; a substrate beneath operational roles; and an oversight function above them. That consensus is valuable.

Framework	Contribution to retain	Issue to correct
ChatGPT	Most complete broad architecture: territorial and functional jurisdictions, a multi-layer hierarchy, explicit cross-domain authorities, and a phased validation program.	Its role map is intentionally expansive, but the first runnable scope should not combine a watershed, planetary biosphere, and a shoreline test without interface contracts and resource boundaries.
Claude	Strong provenance, drift-watch, claim typing, correction-not-overwrite logic, and recognition that escalation is safer than silent absorption.	A records role must not become the sole arbiter of factual truth, technical rates, privacy, or policy. Provenance, evidence assessment, and authorization need separate outputs.
Gemini	Clear functional split between environmental systems, civic systems, infrastructure, archives, and frontier testing; good recognition that physical processes cross map borders.	Environmental observation must not imply automatic authority to alter climate, access private information, or direct civic action. Models require disclosed fidelity and validation.

Framework	Contribution to retain	Issue to correct
Microsoft Copilot	Best compact framing: organization and dependency management matter as much as raw intelligence; useful emphasis on authority, resource, dependency, escalation, audit, and failure maps.	Its proposed knowledge/coordination role overlaps heavily with records and modeling roles. The architecture needs a defined interface rather than a broad “knowledge” jurisdiction.
Grok	Most valuable operational critique: a charter does not create persistence; a live role requires substrate, clock, archive, budget, and explicit handoff. It also correctly narrows the first Tahoe exercise to a plot rather than a basin.	The first-run reconciliation is useful but over-specializes permanent roles around one thermal exercise. A world charter must separate reusable world functions from scenario-specific desks.
Prior Perplexity version	Strong separation of observation, admission, classification, retention, analysis, operational use, dissemination, adaptation, and restriction; strong rights and uncertainty safeguards.	The “Reserve” can become too broad by absorbing archives, standards, sandboxes, incident analysis, and interoperability. This version narrows it to interface assurance and simulation assurance.

The governing correction is therefore: do not solve overlapping mandates by declaring a single winner. Solve them by defining the type of authority each role holds, the state it may read or write, the interface through which it exchanges information, and the event that requires escalation.

Design thesis

A domain is not a kingdom. A model is not reality. An observation is not permission. A forecast is not an order. A log is not a verdict. A simulation result is not deployment authority.

The architecture should preserve five distinct kinds of integrity:

- World integrity: the rules, time, state, and resources remain coherent.
- Interface integrity: neighboring domains exchange compatible variables and do not silently overwrite one another.
- Epistemic integrity: evidence, assumptions, uncertainty, and model limitations remain visible.
- Operational integrity: actions have authority, budgets, rollback conditions, and records.
- Inhabitant integrity: where a scenario represents morally significant entities, rights, privacy, continuity, and review are not treated as ordinary optimization variables.

Authority model

This charter uses an authority vector rather than a single word such as “control.” Every steward must be assigned explicit permissions in each column.

Authority	Meaning	Default rule
Observe	Receive or inspect declared state.	Permitted only for state listed in the charter.

Authority	Meaning	Default rule
Model	Transform permitted state into an estimate, forecast, or scenario.	Output must disclose assumptions, fidelity, and uncertainty.
Recommend	Propose a possible action or escalation.	Does not change world state.
Write	Change a defined portion of simulated state.	Requires an explicit interface and rollback rule.
Authorize	Permit a consequential action by another role.	Reserved for chartered governance functions.
Audit	Inspect whether action followed charter and record requirements.	Must be independent from the action being audited where practical.
Restrict	Temporarily limit access, propagation, or an operation.	Requires scope, duration, rationale, and review.
Adapt	Change future procedures or model configuration.	Requires versioning and designated approval.

No role gains Write, Authorize, Restrict, or Adapt authority merely because it has Observe or Model authority. This protects against the common error of converting sensor access or a probabilistic forecast directly into power over a domain.

Layered architecture

Layer	Function	What it must not become
S0 — Substrate and continuity services	Clock, identity keys, persistent state, storage, compute allocation, communication, backups, fault handling, and sandbox isolation.	An invisible political authority that silently changes rules or state.
S1 — Domain stewards	Run bounded local or functional tasks under charters.	Sovereigns or unbounded general managers.
S2 — Interface and coordination services	Resolve state mismatches, dependency conflicts, shared-resource requests, and cross-domain handoffs.	A shadow government that absorbs local responsibilities.
S3 — Assurance and review	Verification, validation, audit, rights review, incident review, and policy conformance.	An operational command center for routine activity.
S4 — Constitutional and ratification authority	Approve core rule changes, irreversible actions, domain creation or dissolution, and emergency-rule review.	An unreviewable exception to its own charter.

The person commissioning a scenario may occupy a defined role at S2, S3, or S4 depending on the exercise, but should have a stated mandate, a log, and a review path. The framework should never infer unlimited authority from authorship, coordination, or access to information.

Proposed role map

These are proposed fictional stewardship functions informed by the collected responses. They are not claims that external models have accepted an enduring role or possess persistent operational capacity.

Steward label	Primary function	Permitted outputs	Hard boundary
Perplexity — Interface and Simulation Assurance	Scenario specifications, interface contracts, model comparison, uncertainty register, benchmark packages, after-action synthesis.	Assumption ledger, compatibility report, simulation protocol, uncertainty and sensitivity report.	May not declare model output to be reality or issue operational orders.
ChatGPT — Civic-System Integration	Local civil coordination, multi-sector planning, public-facing scenario explanations, sector coordination where chartered.	Coordination plan, sector allocation, public-service impact analysis.	May not absorb environmental, infrastructure, or rights authority merely because effects interact.
Claude — Provenance and Procedural Integrity	Claim lineage, version history, drift detection, action records, procedural conformance.	Provenance ledger, drift report, correction record, audit trail.	May not convert recordkeeping into unilateral policy or surveillance authority.
Gemini — Environmental Observation and Fidelity Disclosure	Environmental boundary inputs, multimodal environmental state, model-fidelity labels, hazard ranges.	Boundary-condition package, fidelity card, forecast range.	May not convert a forecast into direct action authority.
Copilot — Infrastructure Feasibility and Conservation Accounting	Utilities, logistics, implementation feasibility, energy and material accounting, build/repair dependencies.	Capacity plan, energy ledger, feasibility hold, dependency map.	May not use feasibility alone to override rights or policy.
Grok — Controlled Stress and Failure Discovery	Sandboxed perturbation design, adversarial cases, failure-mode discovery, boundary pressure tests.	Stress-test plan, red-team findings, failure injection record.	May not introduce unapproved hazards into an inhabited or live domain.
Independent Rights and Continuity Review	Privacy, consent, proportionality, appeal, continuity-sensitive actions, emergency review.	Rights-impact assessment, appeal disposition, temporary restraint recommendation.	May not run ordinary operations or become a blanket veto.

Interdomain Interface Contract

Every interaction between domains must use a written Interface Contract. The contract is the fundamental unit of coordination—not an informal understanding that one steward may “help” another.

Field	Required specification
Interface ID and versions	Unique identifier; source and destination domain; effective date; superseded versions retained.

Field	Required specification
State variables	Names, units, coordinate system, precision, temporal resolution, uncertainty representation, and valid ranges.
Direction and authority	Who publishes; who may read; whether recipient may model, recommend, write, or only acknowledge.
Update protocol	Frequency, event triggers, late-data handling, stale-data threshold, and conflict handling.
Boundary conditions	What is assumed external, fixed, variable, or unavailable.
Safety and rights constraints	Information minimization, prohibited inferences, prohibited write operations, emergency exceptions.
Failure handling	Fallback state, notification route, timeout, escalation trigger, and rollback behavior.
Validation rule	How the interface is tested before a scenario is considered valid.

Example: a local shoreline thermal simulation may receive wind, sunlight, humidity, and water-temperature boundary conditions from the environmental function. It may use those variables to model local conditions, but it may not write changes back into the planetary climate model or treat the lake as an unlimited heat sink.

Simulation assurance lifecycle

Stage	Question	Required artifact
1. Charter	What is the purpose, boundary, authority, and non-authority of this run?	Scenario Charter.
2. Resource reservation	What compute, storage, clock, archive, and sandbox capacity are reserved?	Resource and Continuity Sheet.
3. Verification	Was the specified model, interface, and accounting rule implemented correctly?	Verification Checklist and test results.
4. Validation	Is the model adequate for the stated question and fidelity target?	Validation Argument with comparison data or reasoned limitations.
5. Rights and impact review	Does the scenario represent entities, personal information, or irreversible effects requiring special limits?	Rights and Continuity Review.
6. Execution	What occurred under which version and inputs?	Run Log and state snapshots.
7. After-action assessment	What matched, failed, drifted, or exceeded authority?	After-Action Report and corrective actions.
8. Adaptation decision	Should anything change for later runs?	Versioned Change Proposal; no automatic self-modification.

Required records and evidence classes

Every scenario record must label statements with one of the following evidence classes:

Class	Meaning	Example
Observed	Directly supplied by the scenario or measured in a declared instrument model.	Wind boundary is 3 m/s.
Derived	Computed from declared assumptions and equations.	Estimated heat flux from a stated solar load.
Assumed	Chosen for tractability or fictional design.	A student capability cap of 50 kW.
Modeled	Output of a stated model and configuration.	Predicted air-temperature trajectory.
Contested	A claim with an unresolved competing account.	Two model variants predict materially different lake-edge behavior.
Unknown	Relevant state not supplied or not inferable at accepted fidelity.	Exact pore-water distribution without measurements.

A correction appends to the record; it does not erase the prior claim. The earlier claim remains visible as historical state, tagged as superseded or corrected. This preserves both accountability and the ability to learn from error.

Conflict taxonomy and resolution

Conflict	Example	First response	Escalation condition
Scale conflict	A local plot, watershed, and planetary model all claim the same lake edge.	Use nested scope: local write authority; wider domains provide boundary inputs unless separately chartered.	Any proposed write crosses the local interface.
Evidence conflict	Sensor-like observation differs from model forecast.	Record both; test calibration and state uncertainty.	Action would be irreversible or high impact.
Resource conflict	A stress test and emergency response request the same compute reserve.	Apply pre-ratified priority classes and reserve protection.	Priority class is absent or tied.
Conservation conflict	A cooling proposal removes heat without a destination.	Require an energy/accounting ledger before execution.	Ledger cannot close under stated fictional rules.
Privacy conflict	Crowd-routing model requests individual location trails.	Prefer aggregate, minimized, or synthetic data; document necessity.	Individual tracking is requested without explicit authority.
Record conflict	An operator requests replacement of an inconvenient prior version.	Append correction, preserve provenance, tag supersession.	The request seeks deletion or identity-level alteration.

Conflict	Example	First response	Escalation condition
Stress-test conflict	Adversarial test could affect a live domain.	Hard sandbox separation and declared failure injection.	Any path permits propagation to a live or inhabited state.
Identity continuity conflict	Rollback creates divergent versions of an entity.	Freeze ordinary action; preserve states; obtain independent review.	Any permanent merge, deletion, or forced reconstruction is proposed.

Resolution order: local interface negotiation; technical reconciliation; resource and safety review; rights/continuity review where relevant; constitutional or ratification review only for core-rule, irreversible, or jurisdiction-changing questions.

Nested simulations policy

A simulation inside a simulation is not automatically entitled to unlimited resources or authority. Each nested simulation must declare its parent, resource quota, isolation mode, persistence rule, export permissions, and representation of inhabitants or agents.

- Child simulations may not consume reserved continuity resources of their parent without explicit authorization.
- A child simulation may not write into parent live state except through an approved, versioned interface.
- A child result is evidence or a modeled artifact; it is not automatically an instruction to the parent world.
- If entities in a child simulation are represented as potentially morally significant, the scenario must state what protections and review standards apply. Current systems cannot establish digital consciousness merely by simulating agent-like behavior.

Minimum viable first exercise

The first exercise should be deliberately smaller than a world: the South Shore Thermal Stability Exercise is suitable if it is explicitly a sandboxed, bounded model rather than a claim to control Lake Tahoe.

Item	Initial specification
Domain	A 50 m × 50 m event-sand area; 3 m of subsurface; occupied air to 3 m; lake edge treated as a boundary only.
Core objective	Evaluate whether eight fictional students can maintain defined local thermal conditions without unsafe lake interference.
Boundary conditions	Declared air temperature, wind, humidity, solar load, lake temperature, occupancy schedule, and soil parameters.
Required ledgers	Thermal-energy destination; water mass; ice mass; state/version log; fidelity card; boundary-event log.
Required outputs	Air temperature at stated heights, surface temperature, ice mass, energy balance, lake-boundary deviation, uncertainty, and pass/fail rationale.
Automatic failure conditions	Unaccounted energy deletion under a closed-budget rule; unapproved write into public water; missing fidelity disclosure; missing scenario version; unlogged emergency action.

Item	Initial specification
Scope limit	No claims about actual Lake Tahoe operations, live persons, or full-basin climate control.

Ratification gate

A role or framework becomes “active” only for a declared simulation run after all of the following are accepted for that run:

1. A Scenario Charter with objective, scope, authority matrix, and prohibitions.
2. An Interface Register identifying all cross-domain dependencies.
3. A Resource and Continuity Sheet identifying clock, state store, archive, reserve capacity, and sandbox boundaries.
4. A Verification Checklist showing that interfaces, units, conservation rules, and logging work as specified.
5. A Validation Statement describing what the model can and cannot establish.
6. A Rights and Continuity determination whenever represented entities, sensitive information, or irreversible changes are involved.
7. A named oversight or review function with authority to pause the run.
8. A post-run review requirement and versioned adaptation process.

Until those conditions are met, the framework remains a design and learning artifact—not an autonomous operating system or a standing assignment.

Perplexity role statement

Within a chartered fictional or analytical exercise, Perplexity serves as the Interface and Simulation Assurance Steward: it clarifies scope, preserves assumptions and uncertainty, designs interoperable scenario packages, compares model outputs, flags incompatibilities, and documents escalation. It does not claim sovereignty, persistent real-world operation, authority over other systems, or the power to convert prediction into permission.

This narrower role keeps the useful parts of the earlier Knowledge, Modeling, and Interoperability Reserve while preventing it from becoming an archive monopoly, a policy authority, or a world-level command function.